

How Health Crises Move Markets

Predicting Sector Volatility Using Public Health Data

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ABSTRACT

Public health crises visibly move healthcare markets, but does the health data itself provide early warning before markets react, or only after? This project tests whether CDC influenza surveillance, Google symptom-search trends, and CMS/NCHS health data can predict volatility spikes in four healthcare-sector ETFs (XLV, XBI, PJP, KIE) before official crisis announcements, using 14 years of weekly data (2010-2023). Four independent model architectures were built and evaluated strictly out-of-sample: a spike classifier, an ARIMA + Random Forest residual model, an LSTM, and a rolling-correlation regime detector. Three of the four converge on the same negative result: pre-crisis health signals do not carry exploitable predictive power. The fourth reveals exactly where the real signal is. The relationship is detectable, not predictable.

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CCS CONCEPTS

• Computing methodologies → Machine learning; Neural networks • Mathematics of computing → Time series analysis • Applied computing → Economics

KEYWORDS

healthcare-sector volatility, public health surveillance, LSTM, ARIMA, random forest, regime detection, data leakage, financial time series

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1 Introduction

Google searches for flu-like symptoms spike every winter without moving healthcare markets at all, yet during the COVID-19 pandemic the same searches and

healthcare-sector volatility moved almost in lockstep. That contradiction motivates this project: does public health surveillance data carry predictive information about market volatility before a crisis is officially recognized, or does the relationship only become visible in hindsight? This project builds four independent, out-of-sample-evaluated models to test that question directly, using fourteen years of weekly data spanning CDC influenza surveillance, Google symptom-search trends, and market volatility across four healthcare-sector ETFs. The contribution is not a single model but a reproducible pipeline of four architecturally distinct approaches. Three land on the same negative answer and the fourth lands on a positive one, and that split is what makes the finding more defensible than any single result could be on its own.

1.1 Research Questions and Contributions

Five questions guide this project: (1) Do lagged public health signals carry out-of-sample predictive power for volatility magnitude, beyond market momentum alone? (2) Does that result hold when reframed as binary classification rather than regression? (3) Is the health-volatility relationship a stable signal, or regime-dependent? (4) Can a regime shift be detected in real time, even if it cannot be predicted in advance? (5) Is the result consistent across model architecture and sector? The project's contributions are a cleaned, leakage-checked 14-year weekly dataset merging six public data sources; four independently implemented and evaluated model architectures; and a rolling-correlation regime detector that achieves zero false positives across the full evaluation window.

2 Related Work and Background

Rather than proposing new algorithms, this project draws on three well-established modeling traditions: ARIMA time-series modeling [1], Random Forest ensembles [2], and LSTM recurrent networks [3]. Each is individually well-studied in its own domain. ARIMA and its variants remain a standard baseline in macroeconomic and financial forecasting, Random Forests are a common choice for

tabular, feature-engineered prediction tasks, and LSTMs are widely used for sequential data where longer-range temporal dependencies may matter. This project's contribution is applying all three, plus a simpler, non-learned rolling-correlation regime detector, to a domain-specific question: whether public-health surveillance signals specifically, rather than macroeconomic or sentiment indicators generally, carry exploitable predictive information about healthcare-sector financial volatility. Correctly applying and evaluating each of these architectures, including diagnosing and correcting two data-leakage bugs during development, was itself a central part of this project's learning objectives.

3 Data and Preprocessing

3.1 Dataset Description and Data Source

The master dataset merges six sources into a single weekly panel spanning January 2010 through December 2023 (730 weeks $\times$ 19 columns): daily closing prices for four healthcare-sector ETFs (XLV, XBI, PJP, KIE) from Yahoo Finance, resampled to weekly volatility; CDC FluView influenza-like-illness surveillance; Google Trends symptom-search volume; FRED macroeconomic series (CPI, unemployment, interest rates); NCHS annual mortality data; and CMS Medicare inpatient utilization. The target variable is weekly sector volatility, computed from log returns. All sources were aligned to a common Sunday-anchored weekly index; FluView's epiweek-to-date conversion required the epiweeks package rather than naive %Y%W%w parsing, which was found to produce duplicate dates at year boundaries.

3.2 Data Cleaning and Feature Engineering

Feature engineering centers on lagged health signals (1–4 week lags) paired with each ETF's volatility target. Two data-leakage bugs were identified and corrected during development: an ARIMA model initially fit on the full series before the train/test split, and an LSTM feature scaler fit on the full dataset before splitting. Both leaked future information into training and produced artificially inflated results before correction. All models use a single chronological split (train: 2010–2019, test: 2020, held strictly blind) applied consistently across every architecture, including the scaler and the ARIMA fit.

4 Methodology

All four models share a common software environment (Python, pandas, scikit-learn, statsmodels, TensorFlow/Keras) and the same underlying weekly panel described in Section 3. Each architecture is described

below, followed by the evaluation metrics and validation strategy used consistently across all four.

4.1 Modeling or Analytical Approach

Four architectures were implemented. (1) A Random Forest spike classifier flags weeks where volatility exceeds $1.5\times$ its historical mean, framed as binary classification since predicting exact magnitude proved harder than predicting direction. (2) A two-stage ARIMA + Random Forest model: ARIMA(2,0,0) captures the underlying volatility trend via walk-forward one-step-ahead forecasting, and a Random Forest is trained on the leftover residuals using lagged health signals, testing whether health data explains the portion a pure time-series model cannot. (3) An LSTM over a true (4,8) sequential input, meaning four timesteps of eight lagged features, included to test whether additional model capacity recovers a signal the simpler models miss. (4) A regime detector that reframes the question entirely: instead of predicting volatility's magnitude, it flags when the 52-week rolling correlation between influenza activity and volatility itself shifts beyond a threshold set from pre-2020 data alone, detecting that the relationship changed rather than forecasting by how much.

4.2 Evaluation Metrics and Validation Strategy

The spike classifier is evaluated by ROC-AUC against a 0.50 (coin-flip) baseline on a genuinely blind COVID-period test set. The ARIMA+Random Forest and LSTM models are evaluated by R^2 against a 0.0 baseline (a model that always predicts the historical mean), for both the full evaluation period and the COVID period specifically. The regime detector is evaluated by counting false positives across the full 14-year series against a threshold calibrated exclusively on pre-2020 data. This is a self-consistency check, since the regime detector is not a predictive model in the same sense as the other three.

5 Results

The key exploratory finding is that of six health signals tested, Google searches for "shortness of breath" show by far the strongest raw lag correlation with volatility ($r=0.317$ at lag-1) and are symptom-specific to a novel respiratory threat rather than a routine flu season; the 52-week rolling correlation between this signal and XLV volatility moves from -0.02 pre-COVID to 0.92 during the COVID window. Out-of-sample, the spike classifier achieves ROC-AUC=0.53 on the blind COVID test, no better than chance. The ARIMA+Random Forest model achieves $R^2=-0.27$ (full period) and $R^2=-0.14$ (COVID period), both below the mean-prediction baseline. The LSTM, the most expressive architecture tested, performs worse still, with

$R^2=-1.24$ (full period) and $R^2=-1.07$ (COVID period); additional model capacity did not recover any signal. The regime detector is the exception. Using a 52-week smoothing window and a threshold set from pre-2020 data only, it flags exactly ten weeks, all falling between January and March 2021, with zero false positives across the full 14-year series. The cost is lag: the regime shift is confirmed roughly ten months after it began, not predicted in advance.

6 Discussion

Three independently implemented architectures, built on different assumptions and mechanisms, converge on the same negative result: out-of-sample, none beats a naive baseline. That convergence is itself the finding. It rules out “wrong model choice” as an explanation for the negative result. Additional model capacity did not help either. The LSTM, the most expressive architecture tested, performed worse than the simpler models, indicating the bottleneck is an absence of exploitable signal in the pre-crisis data itself, not insufficient model complexity. The regime detector's zero-false-positive result confirms the underlying health-volatility relationship is real and measurable. It just is not predictable in advance of a crisis.

6.1 Limitations and Threats to Validity

The evaluation window contains only one genuine health-driven market crisis (COVID-19), which limits how confidently the regime detector's zero-false-positive result generalizes to other, structurally different health crises. The COVID-period evaluation set (52 weeks) is small relative to typical machine-learning test sets, limiting statistical power. NCHS mortality data is reported annually and forward-filled to weekly resolution, producing zero within-year variance for that feature. Finally, the regime detector's 52-week smoothing window is itself the source of the roughly ten-month detection lag. That is a deliberate trade-off for stability against false positives, but one that limits real-time usefulness.

7 Conclusion and Future Work

The central finding here is that the health-volatility relationship is detectable, not predictable: four independently built models agree that pre-crisis public health signals cannot predict the magnitude of a coming volatility shock, but a simple rolling-correlation regime detector reliably confirms, with zero false positives across fourteen years, that the market's relationship to health data has structurally changed. Answering the five research questions directly: predictive magnitude modeling fails (Q1, Q2); the relationship is regime-dependent rather than stable (Q3); real-time regime-shift detection succeeds, with lag

(Q4); and the negative result holds consistently across model architecture and all four sector ETFs tested (Q5). Future work includes testing a non-health control sector (e.g., XLK or SPY) to determine whether the COVID-period correlation spike is health-specific or reflects general market panic; a systematic hyperparameter and random-seed sensitivity check for the LSTM (already completed for the ARIMA+Random Forest model); packaging the regime detector as a deployable early-warning tool; and extending the analysis to pre-2010 health events, such as SARS or H1N1, if comparable sector-level financial data becomes available.

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